Intelligent Credit Risk Analysis for Accurate and Explainable Loan Applicant Classification

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Abstract—Credit-risk classification demands consideration of unequal error costs on top of predictive accuracy and interpretability. This research investigates how decision thresholds derived during training influence the detection of high-risk applicants and whether cost-based model selection extends effectively to new, unseen applicants. The UCI German Credit dataset, comprising 1,000 records and 20 predictors, is used. Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting models are assessed with an 800/200 stratified split, pipeline-based preprocessing, and five-fold validation during development. Thresholds are chosen to minimize a cost assigning five units per missed bad-risk applicant and one unit per false alarm. Gradient Boosting is selected based on development data at a threshold of 0.13. On the independent holdout set, its recall improves from 0.5500 to 0.8667 and cost reduces from 150 to 122 compared to a threshold of 0.50, although accuracy decreases to 0.5500. Logistic Regression attains the lowest primary holdout cost of 107. Uncertainty is quantified through reproducible bootstrap intervals and Holm-corrected paired comparisons. Sigmoid calibration slightly degrades the Brier score despite lowering cost at its own training-selected threshold. Global SHAP analysis is performed using only the 200 holdout applicants, complemented by local explanations. The findings emphasize the importance of explicitly reporting recall false-positive trade-offs rather than assuming that the best model on development data remains optimal. Limitations include historical benchmark data, limited sample size, and non-nested original tuning, restricting generalizability; FinSight serves as educational decision support. Results backing explicit expression of recall/false-positive tradeoff instead of assuming a development winner is end of story. Benchmark data, sample size, and non-nested original tuning limit generalization: FinSight is education decision support.

Keywords—credit risk, asymmetric cost, threshold selection, uncertainty, SHAP, explainable machine learning

I. INTRODUCTION

Credit-risk models have a straightforward categorization of applicants into good or bad risk, but the costs of misclassification are quite different. A false negative could bring loan losses on repayment, while a false positive may produce an unneeded investigation or access limitation. An accurate model might not be appropriate at the default threshold. Interpretability is also needed for insight into model classification financial applications have relied on feature attribution to examine why models classified a given loan [1]; high-stakes learning makes the case that post-hoc interpretability does not substitute for inherently-interpretable models or meaningful validation [2].

SHAPley Additive explanations (SHAP) Explains prediction on feature contributions [3]. At this stage the explanation supports an assessment of asymmetric cost, not evidence of suitability for lending. The contribution offers a standardizable comparison of four common classifiers investigating discrimination, detection of high risk, threshold choice uncertainty calibration and model behavior. All steps are performed in training pipelines, operating points derived from development data. FinSight supplies the implementation layer for human-centered decision support, no new model or SHAP technique.

The analysis is guided by four research questions. RQ1 investigates how selecting decision thresholds with cost-awareness impacts the detection of high-risk applicants compared to the standard 0.50 threshold. RQ2 explores whether the classifier chosen based on training data with asymmetric misclassification costs continues to be the most cost-effective when applied to unseen data. RQ3 analyzes the trade-offs between accuracy, precision, recall, F1-score, ROC-AUC, and cost as the threshold varies. RQ4 identifies which applicant characteristics have the greatest influence on the selected model, based on SHAP explanations.

II. LITERATURE SURVEY

A. Machine Learning and Explainability in Credit Scoring

Credit-scoring data covers nonlinear prediction and methods to reveal behavior. Castile and Celik analyse explanations for a deep-learning credit model [4]. Becker et al. contrast transparency, auditability, and explainability and show why predictive power, for example, is not enough to justify use of a scoring method [5]. De Lange et al. use a boosted model and SHAP on bank data [6], while Nallakaruppan et al. investigate an explainable credit-evaluation method [7]. Although some of these motivate the analysis of linear and tree models, they use different data and error notions from this benchmark.

B. SHAP and Reliability of Model Explanations

SHAP, a unified measure of additive attribution [3]. Gramegna and Giudici analyze the discriminative content in SHAP and LIME explanations of credit risk [8]. A systematic review discovers wide financial applications and several evaluation problems [9]. Ballegeer et al. review this work and explicitly evaluate explanation stability in instance-dependent cost-sensitive scoring [10]. These three contributions exemplify the reasons why an appealing importance plot is not enough evidence for trustworthy explanations. This study verifies attribution direction and additivity, but not stability across re-trained models.

C. Cost-Sensitive Credit Classification and Threshold Selection

Imbalance complicates both prediction and interpretation. Chen et al. study interpretability for imbalanced credit-scoring data [11]. Li et al. learn thresholds in sequential three-way credit decisions using application-specific losses [12]. Unlike the static 5:1 binary cost ratio used here, their approach involves learning a distribution of operating thresholds, but again emphasizes the importance of mapping operating thresholds to differing decision impact. FinSight reports explicit false-negative and false-positive counts for the separate probability calibration and final binary cutoff.

D. Research Gap

Previous work raises both selection and explanation issues from different angles. The testable empirical question that is of most practical interest is whether the training-based cost-aware selection approach will work on completely independent data, given both its recall/false-positive balance and applicant-level explanations. The paper explores that interaction in a small, reproducible benchmark that explicitly addresses uncertainty and calibration issues, rather than claiming that many other configurations have not been explored.

III. DATA RESOURCES

A. Benchmark and Target Definition

The UCI Statlog German Credit original symbolic dataset [13] has 1,000 records and 20 variables, seven numeric and thirteen categorical fields in the implementation. Variables describe account status, credit history purpose amount, duration savings employment housing age, and other applicant features. Raw target 1 indicates good risk 2 indicates bad risk, pipeline uses to map these to 0 and 1. Bad risk is the positive class, and the fitted classes are [0, 1], inference needs to unravel the probability column by searching for class label 1.

The target distribution includes 700 good-risk and 300 bad-risk applicants, creating a 70:30 class imbalance. If we predict all applicants to be good risk, we get 70% accuracy but do not predict any applicants to be bad risk. This situation demonstrates the importance of metrics like recall, F1-score, ROC-AUC and asymmetric cost besides accuracy. Benchmark cost is a relative penalty not a real estimate of money lost.

B. Data Integrity and Scope

Schema validation verifies row count, column sequence missingness listed category values, and numeric validity. The source checksum for the cache is consistent with the stored model metadata. Stratified sample, with seed=42: dev set comprises 800 records of which 560 are good and 240 are bad, holdout set comprises 200 records of which 140 are good and 60 are bad. Applicant keys and raw/original derived targets are not listed as Predictor variables.

UCI reports coding-information errors in the older data-set and publishes corrected South German Credit data [14]. That older data-set is the same initial source population, not a contemporary external cohort of independent data. A different experiment using corrected-data was studied, but the source download was not available in the execution environment. No corrected-data results are claimed; recording and validation are deferred to another time.

IV. RESEARCH METHODOLOGY

A. Preprocessing and Classifier Comparison

Figure 1 gives an overview of the protocol. The raw split was performed before fitted preprocessing. Numeric fields underwent median imputation while categorical fields performed most-frequent imputation, followed by one-hot encoding (with the addition of special treatment for unseen categories). The LR preprocessing included a standardisation of the numeric predictors. Each transformer was fitted within the associated training pipeline to preclude hold-out data from entering preprocessing.

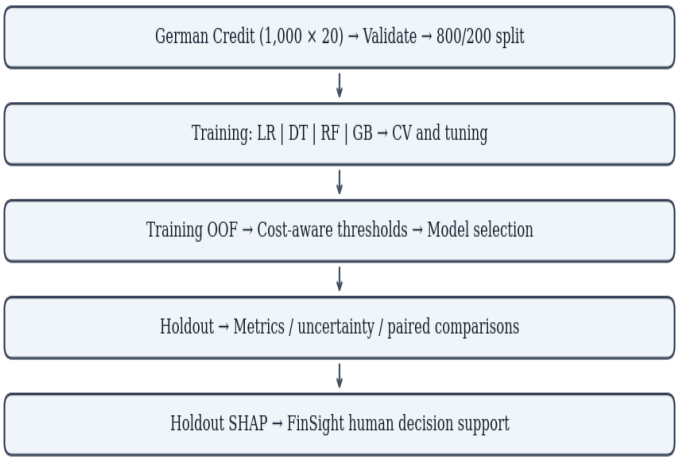


Fig. 1. Compact methodology. All model fitting, tuning, calibration, and threshold selection use development data.

The four models were Logistic Regression (LR), depth-five Decision Tree (DT), Random Forest (RF), and Gradient Boosting (GB). LR, RF, and GB are trained via

8-candidate, randomized searches over the parameters with the greatest five-fold stratified ROC-AUC, while DT serves as an untuned baseline. For LR class weights are not used, the regularization parameter C = 0. 1. RF has 200 trees, depth 5, minimum leaf size 2, feature sampling with square root number of features, and balanced class weights. GB has 200 estimators, learning rate of 0. 05, depth 3, minimum leaf size 5, and subsample of 0. 8. The seeds for all estimators are 42.

B. Operating Threshold and Evaluation Metrics

Five-fold out-of-fold (OOF) development probabilities define each threshold. The selected grid is 0.100.90 in 0.01 steps. The threshold that yields the lowest C = 5FN + FP (discounted equally in cost of an FN or FP) is picked, with tie breakers of greater recall than F1. The model is chosen to minimize the OOF cost, then CV ROC-AUC, then get the holdouts' recall, F1, and stability as tie breakers. The holdouts do not update the model or threshold.

First, Details Accuracy the proportion of correctly classified applicants precision the proportion predicted bad who are actually bad, TP/(TP+FP) recall or Sensitivity the proportion of actual bad applicants, which are correctly classified TP/(TP+FN) F1 the harmonic mean of precision and recall 2TP/(2TP+FP+FN) ROC-AUC -how well the model distinguishes good and bad risk for all thresholds; changing a cutoff will not change the ROC-AUC.

C. Uncertainty, Calibration, and Repeated Validation

For each fixed fitted model and fixed development-selected fixed threshold, 2000 ordinary paired bootstrap resamples of the 200 holdout rows use seed 20261105. Percentile 2.5th and 97.5th quantiles produce 95% confidence intervals. These run conditionally on the fitted pipeline, not while accounting for training/threshold choice variability. Total-cost intervals refer to yet another (200 row) empirical sample; they are not currency intervals nor benchmark estimates at an arbitrarily specified portfolio size.

Six exact two-sided McNemar tests compare paired correctness on the same holdout applicants at each model's chosen threshold. Holm correction adjusts the family of six comparisons at 0.05. Tests compare errors, not ROC-AUC or weighted cost. Calibration employs sigmoid CalibratedClassifierCV with its inner three-fold layering. Five-fold outer OOF predictions of this calibration procedure select its threshold on development data only, then the calibrator is refitted to all 800 development records before holdout evaluation.

Since the initial search and OOF evaluation reuse development folds, the reported estimate is not fully nested. To consider sensitivity to the data split, this additional five-fold, two-repeat stratified validation uses only the 800 development records, without the holdout. Each outer training set estimates a threshold by averaging inner three-fold OOF predictions. This secondary analysis applies only prespecified untuned classifier defaults, and does not revalidate the entire original hyper-parameter search process. Means and standard deviations across ten correlated folds are descriptive only.

D. Explainability and Implementation

Tree SHAP explains the saved GB classifier in bad-risk log-odds. Additivity is verified against its decision function and the logistic transformation verified against class-1 probability. Global importance uses only the 200 holdout applicants and encoded-feature mean absolute attributions are summed by the original predictor. Local explanations preserve signed contributions: positive values raise modeled bad risk and negative values lower it; these positive and negative values describe what the model does, not what causes changes.

Analytics data is stored in S3, schema stored in the Glue Data Catalog, and SQL summaries in the Amazon Athena SQL engine. The catalog table was registered using the Athena DDL, no Glue crawler is needed. These elements support reproducibility and descriptive analytics and are not research novelty. Live cloud access was not repeated for this manuscript.

A Flask app has loaded the persisted pre-processing and model pipeline for inference. Plotly supports portfolio views, while the applicant and what-if pages expose the model output and SHAP factors without refit at request time. The 0100 output is a model calculated educational risk score, which is 100 times the class-1 model output. Its bands of display are project specified and apart from the classification threshold.

V. RESULT AND DISCUSSION

A. Conventional Classification Performance

Table I documents holdout performance at a threshold of 0.50. ROC-AUC is between 0.6741 for the DT and 0.8095 for LR, with GB at 0.8005. The discrimination results do not speak to the desired operating point. For example, GB finds just 33 of the 60 bad-risk applicants at the usual threshold. Fig. 2 shows the holdout ROC curves.

TABLE I. PERFORMANCE AT CONVENTIONAL THRESHOLD 0.50

Model	Acc.	Prec.	Recall	F1	AUC	Cost
LR	0.790	0.714	0.500	0.588	0.809	162
DT	0.650	0.424	0.467	0.444	0.674	198
RF	0.720	0.523	0.750	0.616	0.791	116
GB	0.790	0.688	0.550	0.611	0.800	150

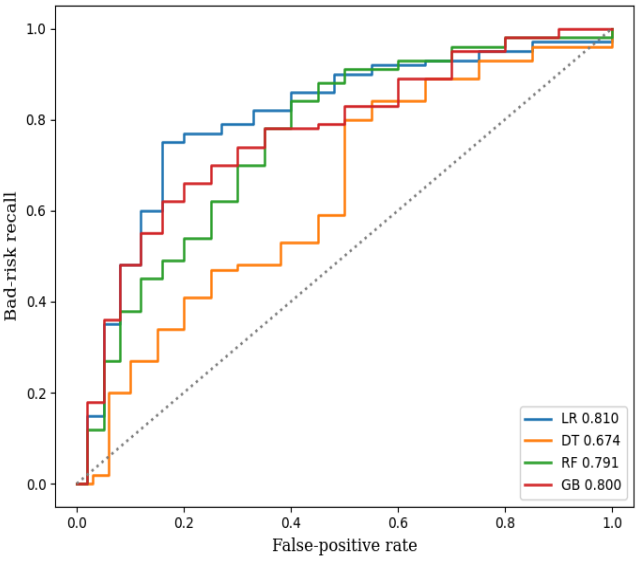


Fig. 2. Holdout ROC curves. Bad risk is the positive class; n = 200.

B. Cost-Aware Threshold Selection

The OOF thresholds are 0.18 (LR), 0.10 (DT), 0.34 (RF) and 0.13 (GB). The development costs are 410 476 416 and 398. GB is selected on the training evidence. The bad-risk recall at its OOF threshold is 0.9125 and the mean CV ROC-AUC over folds is 0.7962 with fold std 0.0367. These are selection-affected development summaries at the selection stage and should not be taken as independent evidence of generalization.

TABLE II. PERFORMANCE AT TRAINING-SELECTED THRESHOLDS

Model	Thr.	Acc.	Prec.	Recall	F1	AUC	Cost
LR	0.18	0.605	0.424	0.883	0.573	0.809	107
DT	0.10	0.605	0.416	0.783	0.543	0.674	131
RF	0.34	0.530	0.384	0.933	0.544	0.791	110
GB	0.13	0.550	0.388	0.867	0.536	0.800	122

C. Independent Holdout Performance

Table II shows the answers to RQ1RQ3. For GB, decreasing the threshold from 0.50 to 0.13, increases recall from 0.5500 to 0.8667, decreases FN from 27 to 8, and decreases cost from 150 to 122 (18.7%). FP is increased from 15 to 82; accuracy is decreased from 0.7900 to 0.5500 and precision from 0.6875 to 0.3881. Fig. 3 shows the confusion matrix for the chosen operating-point: TN 58, FP 82, FN 8, TP 52. This will find more high risk incidents at the penalty of much more review workload..

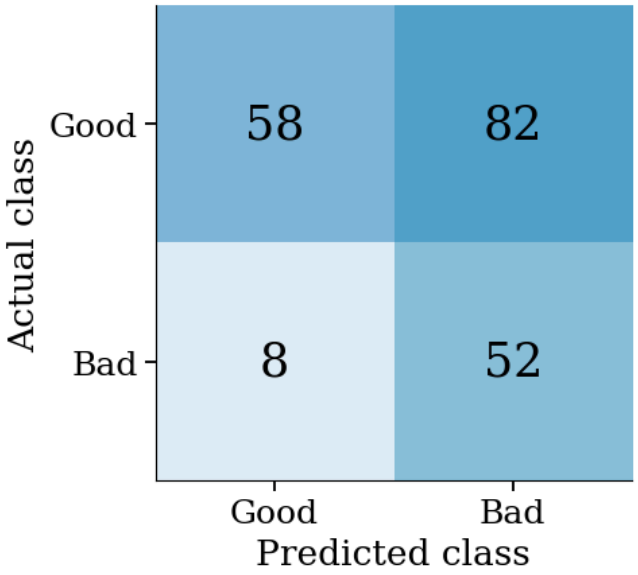


Fig. 3. Selected GB at threshold 0.13. Rows are actual and columns predicted classes.

The training winner is not necessarily the lowest primary holdout cost either. LR obtains cost 107, RF 110, GB 122, and DT 131. LR also outperforms GB for recall 0.8833 on fewer false positives. This reversal negates RQ2 in a fixed split since the retrospective replacement of the chosen model with the test label is not valid. Predicting everyone is good gives cost 300, while predicting everyone is bad gives cost 140. The chosen GB beats both cost baselines, despite its accuracy below the majority class baseline.

D. Threshold Sensitivity Analysis

Figure 4 brings a descriptive evaluation of the implemented threshold range on the holdout set. Lower cutoff values typically boost bad-risk recall but also raise the number of false alarms; conversely, higher cutoffs reduce false alarms but result in a five-unit penalty due to missed bad-risk applicants. The decision tree (DT) exhibits step-like behavior because its leaf probabilities are discrete. These curves illustrate operating sensitivity rather than identifying a new holdout-optimal threshold. The cost ratio and threshold grid are design choices; varying losses or applicant populations may alter the preferred cutoff.

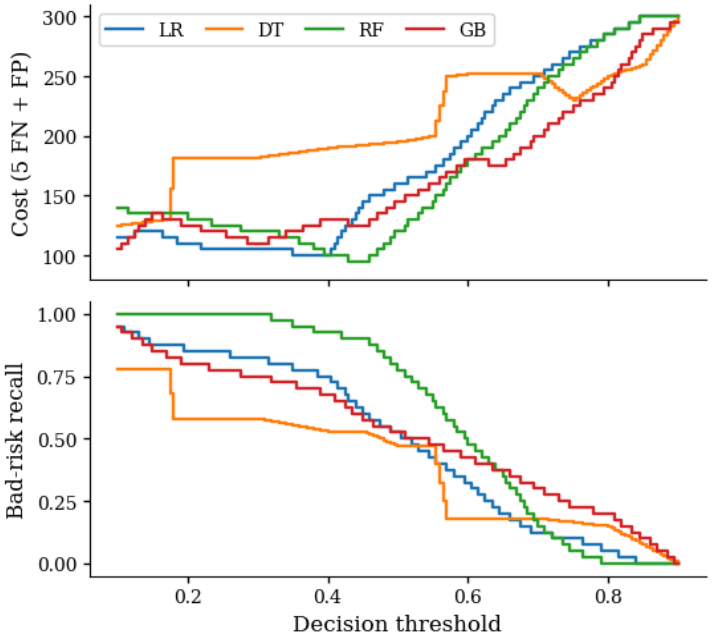


Fig. 4. Holdout cost and recall across thresholds 0.10–0.90. Curves are diagnostic; operational cutoffs were selected on training data.

E. Statistical and Uncertainty Analysis

TABLE III. GB HOLDOUT ESTIMATES AND 95% BOOTSTRAP INTERVALS

Metric	Estimate	95% interval
Accuracy	0.550	0.480–0.620
Precision	0.388	0.306–0.471
Recall	0.867	0.774–0.947
F1	0.536	0.446–0.618
ROC-AUC	0.800	0.723–0.868
Total cost	122.000	95.000–152.000

Figure III contains the uncertainty for the models of the conditions tested. Recall estimates for such few 60 bad-risk holdouts are imprecise, bootstrap intervals not accounting for variability in the model-fitting stage. The broad completed intervals for all four of the models are given plus the reproducibility estimates. Overlapping marginal intervals do not qualify as a paired significance test.

TABLE IV. EXACT McNEMAR TESTS WITH HOLM CORRECTION

Pair	Raw p	Holm p	p < .05
LR–DT	1.0000	1.0000	No
LR–RF	0.0081	0.0488	Yes

LR–GB	0.0433	0.1731	No
DT–RF	0.0315	0.1577	No
DT–GB	0.1439	0.4316	No
RF–GB	0.4807	0.9614	No

All six pairwise comparisons are listed in Table IV. Corrected significant comparisons are LRRF. As they relate to unweighted correctness at the chosen cutoffs, these results cannot demonstrate an edge in asymmetric cost. The low observed cost on this sample is so an empirical finding, not a statistically significant cost benefit.

In the distinct development-only repeated validation, mean cost per applicant (fold standard deviation) is LR 0.562 (0.056), DT 0.623 (0.089), RF 0.546 (0.082), GB 0.550 (0.043). Untuned default settings and correlated folds prevent direct comparison to Table II. This analysis assesses the sensitivity of the simpler pre-specified protocol; an entirely nested evaluation of the full search is still needed.

F. Probability Calibration

The reliability curves shown in Fig. 6 correspond to the Brier score evaluation. Applying sigmoid calibration alters the holdout Brier score from 0.1562 to 0.1569, indicating a slight decline. The development out-of-fold (OOF) Brier score also increases from 0.1644 to 0.1657. The threshold selected during training is 0.20, yielding a holdout ROC-AUC of 0.7993, recall of 0.9000, and cost of 106. In contrast, the uncalibrated model uses a threshold of 0.13, with an ROC-AUC of 0.8005 and cost of 122. A lower cost at a different operating point does not necessarily imply improved probability calibration. That means, the original uncalibrated model remains the primary saved model, and no replacement is selected for deployment based on holdout results.

G. Explainability Analysis

For RQ4, Fig. 5 orders the original predictors by holdout only. Top features: checking account, duration months, purpose credit history, credit amount. Importance is the sum, across the encoded features for each original predictor, of the mean absolute value of the SHAP values. It indicates the contribution strength, not whether the original feature's value acts to increase or decrease risk. Correlated predictors and the tree-path background may dominate attribution.

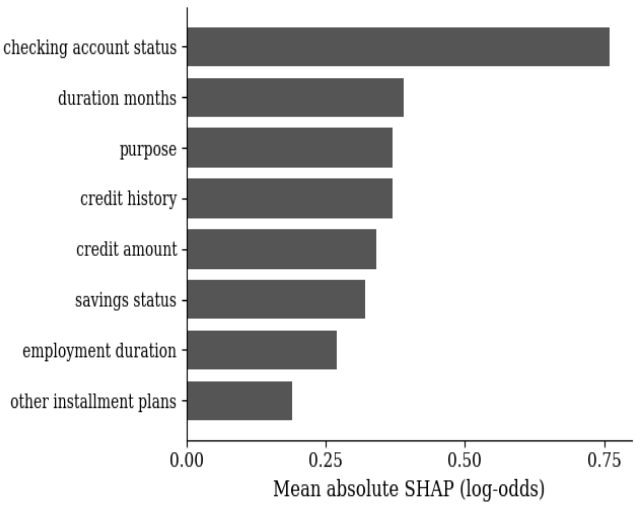


Fig. 5. Global SHAP importance on the 200 independent holdout applicants, aggregated to original predictors.

For one holdout applicant, a risk-increasing feature is checking account status is no checking account (A14) (SHAP +0.309); a risk-decreasing feature is saving status is not.< 100 DM (A61) (SHAP 0.376). Contributions are in bad-risk log-odds and increase model margin, rather than probability; their confirmed signs are true of this particular fitted model. No explanation-stability, causal, or fairness assertion follows from these examples.

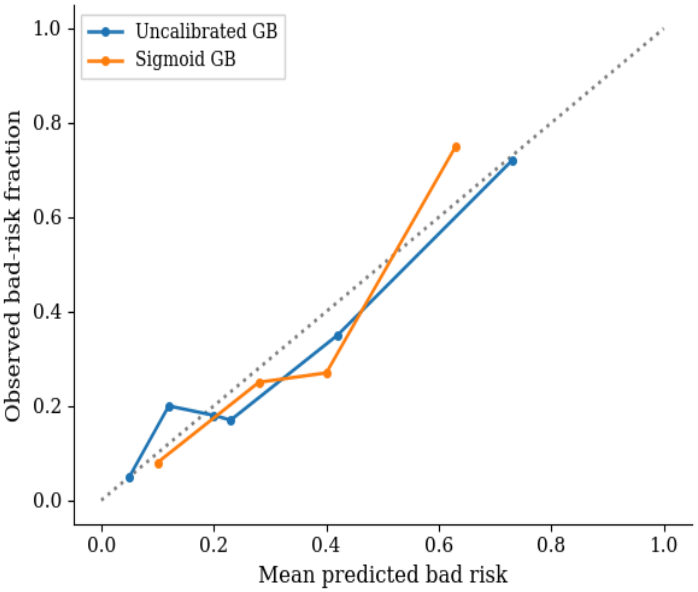


Fig. 6. GB reliability curves using five equal-count holdout bins. The diagonal indicates ideal calibration; n = 200.

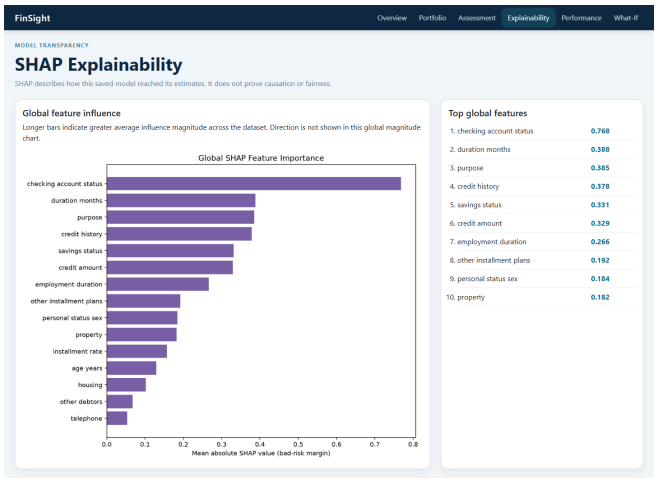


Fig. 7. SHAP-Based Model Explainability Showing Global Feature Importance and Key Credit Risk Predictors.

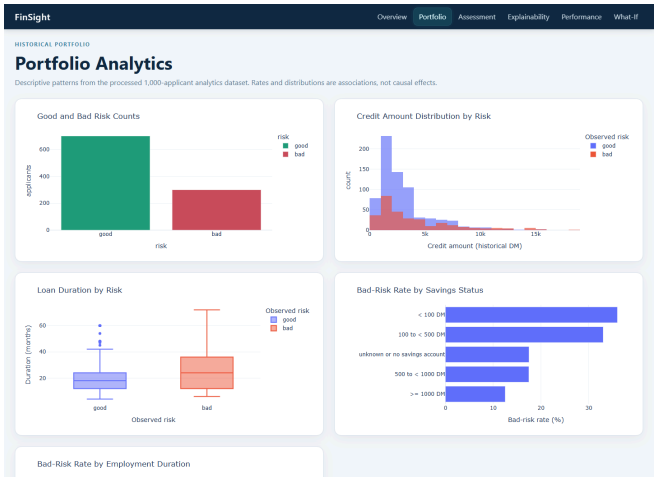


Fig. 8. Portfolio Analytics Dashboard Illustrating Credit Risk Distribution, Loan Characteristics, and Applicant Risk Patterns.

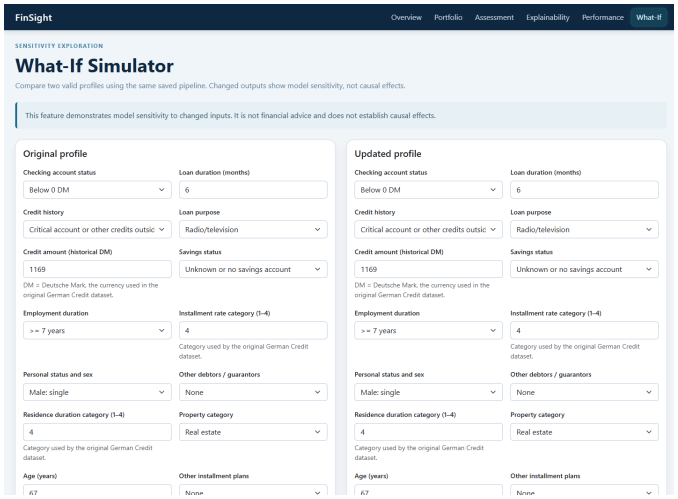


Fig. 9. What-If Simulation Interface for Comparative Credit Risk Analysis.

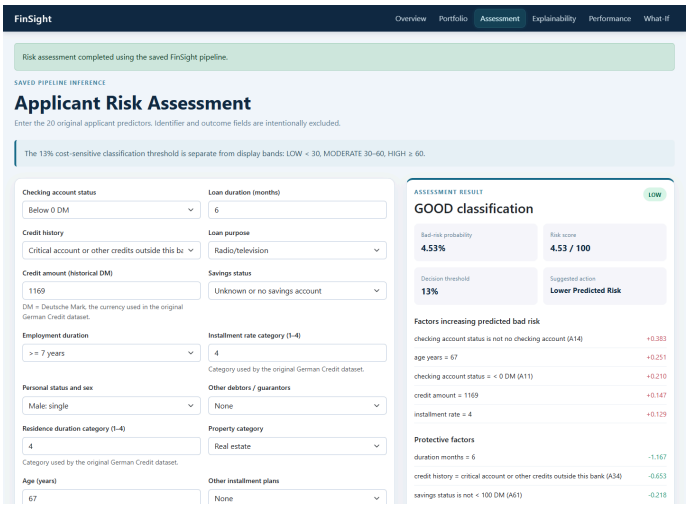


Fig. 10. Applicant Credit Risk Assessment Interface Displaying Predicted Risk Probability, Classification, and SHAP-Based Explanations.

H. System Results and Limitations

All 81 tests in the entire project suite pass without failures; the 50 upstream deprecation warnings still exist. Tests related to data contracts preprocessing inference, dashboard requests, score bounds, and SHAP consistency all completed. The app rest reuse pre-fitted preprocessors and is not training during inference. Cloud support is through implementation and previous validation data, not a new AWS live experiment.

The historical benchmark includes 1,000 records, and the holdout set is limited to 200 rows, which constrains both precision and current applicability. Issues with the original coding and documentation, the presence of sensitive and proxy variables, and the lack of temporal or institution-level external validation limit the scope of interpretation. Bootstrap results depend on the current model fits, while the McNemar test does not assess economic loss. The additional repeated validation does not substitute for fully nested tuning, and no audits of explanation stability or subgroup fairness have been conducted. The fixed cost ratio does not account for loan-specific exposure or opportunity costs. FinSight is not capable of autonomously approving or rejecting actual loan applications.

VI. CONCLUSION

The training-based cost-aware thresholds lead to much greater high-risk detection, but at a cost in false positives. At threshold 0.13, selected GB increases holdout recall from 0.5500 to 0.8667 and decreases cost from 150 to 122 at the price of accuracy dropping to 0.5500. The development-selected model ceases to be the lowest-cost primary classifier on the holdout: LR achieves 107. This highlights the distinction between model selection and independent evaluation.

Bootstrap intervals communicate condition-al test uncertainty, and corrected paired tests limit overstatement of global classification differences. Sigmoid calibration does not lower Brier score, although the threshold based on training data is best in our holdout set. Holdout-only global SHAP and signed local attribution account for the saved model, but do not give certainty about causation. Future investigations should assess the result of independently validated corrected coding, fully nested tuning, an

up-to-date external holdout panel, and the global explanation stability, subgroup convergence and di erences of explained composition. FinSight remains an evidence-educational end-user decision tool; it does not have a credit score or probability of default, does not authorize real underwriting, and is entirely project-tuned.

DATA AND CODE AVAILABILITY

Our benchmark datasets are available in public UCI [13], [14]. The code implementation for preprocessing, model evaluation thresholds explainability, and the FinSight decision-support prototype can be released for academic reproducibility on request.

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